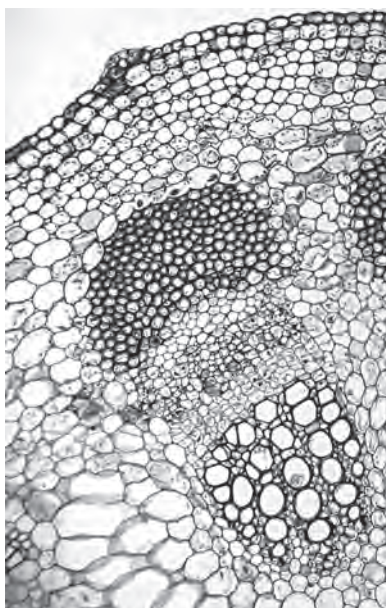


Answer ALL questions.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Plant stems contain phloem, sclerenchyma fibres and xylem vessels.

(a) The photograph shows part of a transverse section through the stem of a plant.



(Source: © Ed Reschke/Getty Images)

(i) Label the photograph to show the position of the phloem, sclerenchyma fibres and xylem vessels.

(3)

(ii) How many of the following statements about phloem are correct?

- connected to other cells by plasmodesmata
- transports water and organic solutes
- translocation moves substances in phloem

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) The cells of plants and prokaryotes have cell walls.

(i) Name a structural molecule in each type of cell wall.

(2)

Plant cell wall

Prokaryotic cell wall

(ii) Explain how the arrangement of molecules contributes to the physical properties of the cell walls of sclerenchyma fibres.

(3)

(Total for Question 1 = 9 marks)

